

ssh bandit#@bandit.labs.overthewire.org -p 2220

N+5th line: sed '#!d' /etc/motd

Password info: grep -n bandit# /etc/passwd

Level: 0

Password: bandit0

5th line Content: . / ;. \ ; ; / /__./\ : |

Password information:37 bandit0:x:11000:11000:bandit level 0:/home/bandit0:/bin/bash

Explanation: The password was given to me on the website

Level 1:

Password: ZjLjTmM6FvvyRnrb2rfNWOZOTa6ip5lf

5th line: . ; / ` ; ' ____ / , .-'. ' \ .

Password info:38:bandit1:x:11001:11001:bandit level 1:/home/bandit1:/bin/bash

Explanation: cat /home/bandit0/readme

Level 2:

Password: 263JGJPfgU6LtdEvvgfWU1XP5yac29mFx

5th line: ; | ; \ ; || : | /__ / \ | ''

Password info:49:bandit2:x:11002:11002:bandit

Explanation: cat /home/bandit1/-

Level 3:

Password: MNk8KNH3Usiio41PRUEoDFPqfxLPISmx

5th line: | : | ; | ' ; | . ; ; \ \ ; :

Password info: 60:bandit3:x:11003:11003:bandit level 3:/home/bandit3:/bin/bash

Explanation: cat /home/bandit2/"spaces in this filename"

Level 4:

Password: 2WmrDFRmJlq3lPxneAaMGhap0pFhF3NJ

5th line: / . : , ' : .-'. ' ;

Password info: 65:bandit4:x:11004:11004:bandit level 4:/home/bandit4:/bin/bash

Explanation: cat ./inhere/...Hiding-From-You

Level 5:

Password: 4oQYVPkxZOOEOO5pTW81FB8j8lxXGUQw

5th line: ' ; \ ; / | ' : ; . \ . \ ;

Password info: 66:bandit5:x:11005:11005:bandit level 5:/home/bandit5:/bin/bash

Explanation: (file -type f | xargs file) then (cat ./-file07)

Level 6:

Password: HWasnPhtq9AVKe0dmk45nxy20cvUa6EG

5th line: \ \ ' , / | | ' \ \ ' \ |

Password info: 67:bandit6:x:11006:11006:bandit level 6:/home/bandit6:/bin/bash

Explanation: found file size using (find . -size 1033c) then (cat ./maybehere07/.file2)

Level: 7

Password: morbNTDkSW6jllUc0ymOdMaLnOIFVAaj

5th line: ; : / ' : | : ' |--"

Password info: 68:bandit7:x:11007:11007:bandit level 7:/home/bandit7:/bin/bash

Explanation: (file -type f -user bandit7 -group bandit6 -size 33c) then I looked for the ones that worked, lastly (cat /var/lib/dpkg/info/bandit7.password)

Level: 8

Password: dfwvzFQi4mU0wfNbFOe9RoWskMLg7eEc

5th line: \ \.' ; |.' \ \;

Password info: 69:bandit8:x:11008:11008:bandit level 8:/home/bandit8:/bin/bash

Explanation: grep "millionth" data.txt (get the line where the text is "millionth")

Level: 9

Password: 4CKMh1JI91bUIZZPXDqGanal4xvAg0JM

5th line: www. `---` ver `---` he `---` ire.org

Password info: 70:bandit9:x:11009:11009:bandit level 9:/home/bandit9:/bin/bash

Explanation: sort data.txt | uniq -c (sort the data and show a count of how many time each line appears)

Level: 10

Password: FGUW5ilLVJrxX9kMYMmIN4MgbpfMiqey

5th line:

Password info: 39:bandit10:x:11010:11010:bandit level 10:/home/bandit10:/bin/bash

Explanation: strings data.txt | grep "=" (gets all the string data and then finds all that have "=")

Level: 11

Password: dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr

5th line:

Password info: 40:bandit11:x:11011:11011:bandit level 11:/home/bandit11:/bin/bash

Explanation: base64 -d data.txt (decodes from base64)

Level: 12

Password: 7x16WNeHli5YklhWsfFlqoognUTyj9Q4

5th line: Welcome to OverTheWire!

Password info: 41:bandit12:x:11012:11012:bandit level 12:/home/bandit12:/bin/bash

Explanation: tr 'A-Za-Z' 'N-ZA-Mn-za-m' < data.txt (shifts the text back 13 letters)

Level: 13

Password: FO5dwFsc0cbaliH0h8J2eUks2vdTDwAn

5th line:

Password info: 42:bandit13:x:11013:11013:bandit level 13:/home/bandit13:/bin/bash

Explanation: Make a temp directory using “mkdir /tmp/(name)”. Cd into the temp directory using “cd /tmp/(name)”. Use “file (name of file)” to check the file. See that it was compressed. Move the compressed file into a decodable file(mv file file.(gz,bz2,tar) depending on the compression technique. Unzip the file using (gzip -d (file name)), (bzip2 -d(fn)), or (tar xf (fn)) depending on the compression technique. Rinse and repeat on the file while removing unnecessary files to keep clean. Cat the file when file type is ASCII.

Level: 14

Password: MU4VWeTyJk8ROof1qqmcBPALh7IDCPvS

5th line: If you find any problems, please report them to the #wargames channel on

Password info: 43:bandit14:x:11014:11014:bandit level 14:/home/bandit14:/bin/bash

Explanation: ssh into bandit 14 using the private key that was given “ssh -i sshkey.private bandit14@localhost -p 2220” then run “cat /etc/bandit_pass/bandit14”

Level: 15

Password: 8xCjnmgoKbGLhHFAZIGE5Tmu4M2tKJQo

5th line: discord or IRC.

Password info: 44:bandit15:x:11015:11015:bandit level 15:/home/bandit15:/bin/bash

Explanation: “nc localhost 30000” then “MU4VWeTyJk8ROof1qqmcBPALh7IDCPvS” use nc (netcat) open a connection to port 30000 as localhost. Then send the password of the previous level through.

Level: 16

Password: kSkvUpMQ7IBYyCM4GBPvCvT1BfWRy0Dx

5th line:

Password info:45:bandit16:x:11016:11016:bandit level 16:/home/bandit16:/bin/bash

Explanation: same as level 15 but use “nc” instead of “nc” and add “--ssl” after “ncat”. So “ncat --ssl localhost 30000” then “8xCjnmgoKbGLhHFAZIGE5Tmu4M2tKJQo”

Level: 17

Password:

-----BEGIN RSA PRIVATE KEY-----

MIIeOglBAAKCAQEAvMokuifmMg6HL2YPIOjon6iWfbp7c3jx34YkYWqUH57SUdyJ
imZzeyGC0gtZPGUjUSxiJSWI/oTqexh+cAMTSMIOJf7+BrJObArnxd9Y7YT2bRPQ
Ja6Lzb558YW3FZI87ORiO+rW4LDCDCNd2IUvLE/GL2GWyuKN0K5iCd5TbtJzEkQTu
DSt2mcNn4rhAL+JFr56o4T6z8WWAW18BR6yGrMq7Q/kALHYW3OekePQAzL0VUYbW
JGTi65CxbCnzc/w4+mQqYvmzpWtMAzJTzAzQxNbkR2MBGySxDLrjg0LWN6sK7wNX
x0YVztz/zbIkPjfkU1jHS+9EbVNj+D1XFOJuaQIDAQABAolBABagpxpM1aolWfvD
KHcj10nqcoBc4oE11aFYQwik7xfW+24pRNUDE6SFthOar69jp5RILwD1NhPx3iBI
J9nOM8OJ0VToum43UOS8YxF8WwhXriYGnc1sskbwpXOUDc9uX4+UESzH22P29ovd
d8WErY0gPxun8pbJLmxkAtWNhpMvfe0050vk9TL5wqbu9AlbssgTcCXkMQnPw9nC
YNN6DDP2lbcBrvgT9YCNL6C+ZKufD52yOQ9qOkwFTEQpjtF4uNtJom+asvlpmS8A
vLY9r60wYSvmZhNqBUrj7lyCtXmLu1kkd4w7F77k+DjHoAXyxcUp1DGL51sOmama
+TOWWgECgYEA8JtPxP0GRJ+IQkX262jM3dElkza8ky5molwUqYdsx0NxHgRRhORT

```
8c8hAuRBb2G82so8vUHK/fur85OEfc9TncnCY2crpoqsgghifKLxrLgtT+qDpfZnx
SatLdt8GfQ85yA7hnWWJ2MxF3NaeSDm75Lsm+tBbAiyC9P2jGRNtMSkCgYEAypHd
HCctNi/FwjulhttFx/rHYKhLidZDFYeiE/v45bN4yFm8x7R/b0iE7KaszX+Exdvt
SghaTdcG0Knyw1bpJVyusavPzpaJMjdJ6tcFhVAbAjm7enClvGCSx+X3l5SiWg0A
R57hJglezliVjv3aGwHwvIZvtszK6zV6oXFAu0ECgYAbjo46T4hyP5tJi93V5HDi
TtieK7xRVxUI+iU7rWkGAXFpMLFteQEsRr7PJ/lemmEY5eTDAFMLy9FL2m9oQWCg
R8VdwSk8r9FGLS+9aKcV5PI/WEKlwgXinB3OhYimtiG2Cg5JCqIZFHxD6MjEGOiU
L8ktHMPvodBwNsSBULpG0QKBgBApItfC1HOnWiMGOU3KPwYWt0O6CdTkMJOML8Ni
blh9elyZ9FsGxsgtRBXRsqXuz7wtsQAgLHxbdLq/ZJQ7YfzOKU4ZxEnabvXnvWkU
YOdjHdSOoKvDQNWu6ucyLRAWFuLSeXw9a/9p7ftpxm0TSgyvmfLF2MIAEwyzRqaM
77pBAoGAMmjmlJdjp+Ez8duyn3ieo36yrttF5NSsJLABxFpdlc1gvtGCWW+9Cq0b
dxviW8+TFVEBI04f7HVm6EpTscDxU+bCXWkfjuRb7Dy9GOtt9JPsX8MBTakzh3
vBgysi/sN3RqRBcGU40fOoZyfAMT8s1m/uYv52O6lgeuZ/ujbjY=
-----END RSA PRIVATE KEY-----
```

5th line: --[Playing the games]--

Password info: 46:bandit17:x:11017:11017:bandit level 17:/home/bandit17:/bin/bash

Explanation: I used “nmap -sV -p 31000-32000 localhost” to show a map of all the open ports along with what service they are using (sV) through ports 31000-32000. The only open on that used ssl/tls was port 31790. This was because all the other ports were servicing other connection methods. Then I sent the previous password and got a private key. I had to save the key to a file in order to use it to ssh into the next level. I also had to make sure that the permissions on for that file were read-only for the owner and nothing for all else.

Level: 18

Password: x2gLTtjFwM0hQ8oWNbMN362QKxfRqGIO

5th line:

Password info: 47:bandit18:x:11018:11018:bandit level 18:/home/bandit18:/bin/bash

Explanation: ssh into the bandit17 server using the private key, “ssh -i key bandit17@bandit.labs.overthewire.org -p 2220”. “diff passwords.new passwords.old” (find the difference between the two).

Level: 19

Password: cGWpMaKXVwDUNgPAVJbWYuGHVn9zl3j8

5th line: This machine might hold several wargames.

Password info: 48:bandit19:x:11019:11019:bandit level 19:/home/bandit19:/bin/bash

Explanation: Had to use “ssh -T”(gets into the shell but doesn’t allow interaction with the shell itself, so the shell can’t kick the user out). No signs of being logged in showed up until I typed “ls” when the “readme” file showed up. “Cat readme” and got the password.

Level: 20

Password: 0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbYO

5th line: If you are playing "somegame", then:

Password info: 50:bandit20:x:11020:11020:bandit level 20:/home/bandit20:/bin/bash

Explanation: “./bandit20-do cat /etc/bandit_pass/bandit20” use “./bandit20-do” as the user and display the password located at “ /etc/bandit_pass/bandit20”. Add “/bandit20” to find the password of the level.

Level: 21

Password: EeoULMCra2q0dSkYj561DX7s1CpBuOBt

Explanation: Create 2 instances of bandit20. Have one instance listen for any connection on a specified port. Have that instance send the password of the level to any connection.

“echo -n '0qXahG8ZjOVMN9Ghs7iOWsCfZyXOUbYO' | nc -l -p 1234” Have the second instance connect to the same port using the setuid of the level “suconnect” (“ ./suconnect 1234”). The second instance receives the current password as soon as it connects and sends the next password back to the first instance.

Time taken:

Around 10 hours

Code for part 2:

“cat King_Lear | tr -d '[:punct:]' | tr '[:upper:]' '[:lower:]' | tr -s ' ' '\n' | sort | uniq -c | sort -n”

```
67 who
69 how
70 king
73 man
73 reg
75 we
75 well
76 know
76 let
77 ill
78 now
79 edn
79 when
80 more
80 which
81 enter
82 or
84 good
85 th
87 am
88 come
90 from
97 lord
98 edg
98 shall
101 on
103 o
104 all
107 by
109 do
110 if
114 sir
115 our
118 glou
119 fool
125 will
127 as
132 but
136 are
137 what
139 thee
140 her
146 so
155 no
158 thy
167 for
172 be
173 kent
174 he
191 it
198 him
207 have
209 his
211 with
214 thou
227 is
227 me
228 your
229 lear
242 this
285 not
298 in
348 that
404 a
454 you
458 my
494 of
558 to
650 i
740 and
853 the
```

Citations:

Terminal: man (ncat, nc, tar, bzip2, gzip, ssh, grep, mv, awk, xxd, chmod, nmap, echo, base64)

ChatGPT:

- How do you make a temporary directory
 - Mkdir /tmp/temp
 - Showed me how to make a temporary directory
- How to you send a message through a certain port in the terminal
 - Showed me how to listen for a connection on a certain port and connect to a port
- How do you ssh into a shell with a private key
 - Told me to make a file with the key inside
 - Got that I need to take away permissions from the file through the error messages
- How do you unzip a file using gzip
 - Told me to move the file into a file with the corresponding ending (.gz)

Documentation:

- All of the sites that OverTheWire had on their levels pages

Collaborators:

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